Timeline: Adding the time dimension to spreadsheets – Revisions cover letter

Dear Reviewers and Program Committee,

Thank you for the detailed and constructive feedback and for encouraging reviews of the paper. We have revised the paper, addressing the main points as outlined in the Author's Response as well as the more specific detailed points raised in the individual reviews.

Overall, the framing of the paper has evolved and it now presents a more comprehensive discussion of the design – rather than overemphasizing the programming language theory perspective, the paper now discusses motivation for the design (small-scale formative study) and also critical evaluation of the system (structured using the cognitive dimensions of notations framework).

The major changes in the revised version are:

1) **Editing formulas with holes has been dropped** (formerly section 3.5).
2) **Discussion of the small-scale formative study has been added** (section 5). This provides details about the study setup, participants, modelling task that we asked participants to solve and findings from the study. The most notable point (two approaches to embedding two-dimensional data into a single dimension) is discussed in more detail.
3) **Case study assessments for the individual case studies have been revised** (now section 6) to explain what each case study adds, why it has been selected and whether it could be implemented using ordinary spreadsheet system (with reference to lessons from the formative study).
4) **Discussion section has been added** (section 7). This first reviews how the two dimensions are used in the case studies (with reference to the formative study findings) and then evaluates usability of Timeline (including visibility and immediacy). This draws on existing heuristic evaluation of spreadsheet systems by Nardi and also using the cognitive dimensions framework.

Numerous other smaller improvements to the paper have been made and can be reviewed by viewing the diff. More notable changes addressing specific concerns by the reviewers include:

1) The abstract has been revised to deemphasize programming language theory perspective; similarly, introduction and contributions now provide a broader framing.
2) Design motivation has been moved from section 2 (Overview) and is now in more detail discussed in the stand-alone formative study section (section 5)
3) Smaller improvements in sections 2 and 3 aim to remove confusion about the working of the `=>` operator and the coeffect-based optimisation algorithm (as well as the coeffect typing rules).
4) Motivation for supporting composable data visualisations is now clarified – and is linked to the need to visualise data over time (in response to the visibility and immediacy issues).
5) Smaller clarifications in section 4 aim to clarify multiple aspects of the formal model (stateful vs. stateless cells, reading of the judgements, checking of the preconditions for the properties).
6) Throughout the paper, we addressed specific nits listed by the reviewers (remove "remarkable" and "first-order", explain `prev(prev(counter))`, check links, fix lemma references).

We believe that the revisions have significantly improved the paper. In particular, we thank the reviewers for their encouragement to move from a narrower programming language theory framing to a more comprehensive design discussion – which we believe serves the system much better!

Thank you,
The Authors